CS 160-01: Introduction to Algorithms (Spring 2025)

CS 160:Introduction to Algorithms

[Go to CS 160 schedule]

General Information

- **Instructor:** Karen Edwards
- **Lectures:** MW 9:00-10:15 (R+ block) in Barnum LL08
- **Attendance:** There will be a lecture feedback sheet approximately once a week in lecture to measure understanding and participation.
- **Office Hours:** See Piazza.
- **Exams:** Dates can also be found on the schedule page.
  - Exam 1 and Exam 2: 75 minutes during regular class. Dates: Wed Feb 19 and Wed Mar 26.
  - Final Exam: CONFIRMED FOR Wed, May 7, from 7-9 PM, in Barnum LL08
- **Homework:** The homework, typically covering the Tuesday and Thursday lecture and the Thursday-Friday recitation is due by 11:59pm on the following Tuesday.
- **Canvas:** Canvas is being used **only** to share video recordings of lectures. There are no guarantees about these videos. If you have to miss a class, a better approach is to rely upon friends with good note-taking skills.

Quick Links:

Prerequisites

The prerequisites for this class are CS 15 and (CS/MATH 61 or MATH 65). If you have not successfully completed both courses either at Tufts or elsewhere and if you are enrolled in CS 160, please contact your instructor. We will be glad to help with references for you to study, but please understand that learning the prerequisite topics is mainly your responsibility.

This class also assumes familiarity with a few math topics such as logarithms, manipulation of summation notation, and summation of various geometric series. Be prepared to brush up on such topics as needed.

Course Logistics

Syllabus

- This page (and the schedule page) function as the course's syllabus.

Textbook

- The course is heavily based on the following book:
  [CLRS] T. H. Cormen, C. L. Leiserson, R. L. Rivest, and C. Stein. *Introduction to Algorithms (4th edition).* MIT Press, 2022.
  - We will also provide section numbers from the 3rd edition, if the topic is in there. (If there is something covered in the new edition that is not there in the older one, or is presented in a different way, be aware that we will be using that newest edition, and your experience is likely to be better if you can gain access to that one.)
  - More info at MIT press. Many students find they can learn more easily when they have access to a hard copy (even if it is an older edition).
  - You may also use this link to access a digital copy of the 3rd edition from Tufts.
  - Two copies of the 3rd edition have been permanently reserved on reserve in the Tisch library for students to access.
  - **Note:** The book is a comprehensive introduction to algorithms and contains many wonderful topics beyond what will be covered in this course.

- For further reading, any of these additional texts is recommended as an extra reference (all of them should also be available in the Tisch library, please alert us if not).
  Several students in the past have found the Skiena text particularly readable and have enjoyed his particular perspective, and yet each of these supplemental texts has had fans!
  - [KT] J. Kleinberg, E. Tardos. *Algorithm Design,* Addison-Wesley, 2005.
  - [DPV] S. Dasgupta, C.H. Papadimitriou, and U.V. Vazirani. *Algorithms,* McGraw-Hill, 2006.
  - [S] S. Skiena, *The Algorithm Design Manual,* (2nd edition) Springer, 2010.
    - Use this link for an online version from Tufts of the second edition. See here for errata.
  - [BVG] S. Baase and A. Van Gelder, *Computer Algorithms*, 3rd edition, Addison-Wesley, Reading, MA, 2000.

Course Tools

- We will use Piazza for class discussions, announcements, polls, handouts, assignments, and so on.
- Homework will be submitted and graded using Gradescope.
- Video recordings of lectures (if successful, no guarantees) will be available on Canvas.

Instructional Team:

Lead Instructor:

- Karen Edwards

Grad TAs:

- Jacob Boerma
- Jocelyn Garcia

Teaching Fellows:

- Dan Patterson
- Kimaya Wijeratna

CAs:

- Daniel Peng
- Eli Morton
- Johnny Tan
- Kevin Yu
- Lakshita Jain
- Marlon Fagundes Pereira Junior
- Megan Yi
- Meghan Kelly
- Merwan Malakapalli
- Peter Scully
- Rafeed Anwar
- Saška Barancikova
- Shayne Sidman
- Stephen Burchfield

Office Hours:

- Office hours are a great place to hang out and get work done, whether or not you have any questions yet. We have a large pool of motivated and talented TAs. Please don't be afraid to join them in office hours as needed to help you further understand the material.
- List of office hours is available on Piazza. Any cancellations, one-time location changes or additional hours will also be noted there.

Expected Work

Participation: Classes, Recitations, Quizzes/Feedback Forms:

- Students are expected to attend class and complete the related reading assignments.
- Students are expected to attend and participate actively in recitation sessions (attendance will be taken; we will drop two). These sessions provide an invaluable opportunity to practice the material and solidify learning.

- There will be lecture feedback sheets approximately once a week in lecture to measure attendance and understanding. We will drop two. Part of your grade will be based on this participation.

Homework:

- Most weeks, you will be given an assignment to practice the material presented in class and discussed in recitation. Due 11:59pm on Tuesday.
- Individual completion and regular timely submission of the homework with full attribution of sources is a prerequisite for passing this course.
- To obtain full credit, you must justify your answers. When describing an algorithm, do not forget to analyse its running time and justify why the algorithm is correct.

Collaboration and Integrity on Homework Submissions

Although you may discuss these problems in the preliminary stages with others, **submitted work should be done individually** and written in **your own words**. If you have any discussions with others, whether students, friends, TAs or faculty, relative to a homework problem or if you gain information from a written source (e.g., website) other than your own notes from lecture or the textbook for the course, you **must identify your collaborator(s)/source(s)** in writing on your homework submission. Failure to cite your sources constitutes an academic integrity violation and may be reported to the Dean.

How to write proofs:

On Piazza there is a short summary containing the key points on how to write proofs. In addition to our guide, there are various materials on the web to check out. We encourage you to look around for those resources whenever you have time in the beginning of the semester.

How to submit homework

- Homework assignments will be available on Piazza. Homeworks are turned in via Gradescope.
- Homeworks **must be typeset** using your preferred software -- LaTex is one option, and it is highly recommended. It is also required for the first homework. In general, you may handwrite or draw diagrams, trees, and graphs, which you must include/embed in the PDF. Any associated math, analysis, or descriptions of the diagrams should be typeset.

Late homework policy:

- You are allowed **~~four (4) now six (6)~~ now eight (8) "tokens"** to be used at your discretion. Each token accounts for a **24-hour automatic extension** on the homework.
- Up to **two (2) tokens** can be used on the same assignment (for a 48-hour maximum extension). After that, if you want to use more than 2 tokens for a longer extension, you need to ask either in Piazza (to "instructors") or in office hours.
- Tokens will be applied automatically based on the Gradescope timestamp of your last submission. You do not need to notify staff that you are using tokens.
- Make sure to keep track of how many tokens you have used. (You can see late submissions on Gradescope).
- Our calculations have a small grace period (of roughly 10 minutes) to account for last minute changes and/or differences in clocks. This grace period is automatically applied to all submissions.
- If you run out of tokens (or submit homework more than 48 hours late), late homework will be counted as 0. However ....
- At the end of the semester we will calculate how many tokens are used by each student. If a student missed the deadline once by a couple of minutes (even after taking into account the tokens and the

grace period), then we will be lenient. Any other case will count as 0, which will be applied to the lowest late homework(s) to minimize impact. So it is always worth turning in homeworks even if you are late and already used all your tokens.
    - In short, be punctual and you will save all of us from troubles! If you have a valid reason for not submitting homework or need to request a longer extension, you should notify your Dean (and, if applicable, Health Services). Decisions on any extension requests will be made in consultation with your Dean.

Exams:

- The midterm exams and the final exam will include questions similar to those included on homework assignments or recitation problems and will draw on material covered in lecture and/or in the reading.
- Exam dates and times are listed under General Information and on the schedule page.
- **There are no make-up exams, so check the exam schedule before making travel arrangements.**

Grading

Your final grade percentage will be the **maximum** of the following two formulas:

> **Overall grade = 25%(Homework) + 5%(Participation) + 2x20%(per midterm exam) + 30%(Final exam)**
>
> OR
>
> **Overall grade = 25%(Homework) + 5%(Participation) + 2x15%(per midterm exam) + 40%(Final exam)**

Clarifications:

- No homeworks will be dropped.
- Participation is half recitation attendance, half lecture attendance (as measured by turning in lecture feedback sheets.) Two recitations and two feedback sheets will be dropped, i.e., you can miss up to 2 of each for any reason. We do not distinguish between "excused absences" and other absences; all such requests are covered by dropping 2 for everyone. If you have valid reasons for missing 3 or more recitations, or 3 or more class gradescope quizzes, please email the instructors.
- Exam grades will not be curved individually, but the final overall letter grades of the course will be adjusted as needed. In particular, anyone in this class whose overall average is at or above the median will earn at least a B+.

Regrade requests:

If you believe any part of your work on Gradescope was graded incorrectly, we want you to let us know via a regrade request on Gradescope. Regrade requests should be concise, directed, and respectful. You should point out a specific aspect of the assignment that was graded incorrectly, and explain why in your own words. We also reserve the right to deduct points if we notice additional errors during the regrading. The person who regrades your submission will send you a response. If you have further concerns thereafter, you may contact one of the TFs or Graduate TAs at office hours or through a private Piazza post. Regrades must be submitted within **one week** of release of the grades, unless stated otherwise.

Tips on getting the most out of this course:

- Come to every class prepared. At a minimum, you should review the previous lecture, and you may also benefit from looking over the reading beforehand. Simply showing up will not suffice for your learning.
- If the pace of the lecture feels too fast, consider reading some basic material beforehand, perhaps in one of the auxiliary texts.
- When studying, try to re-derive the solutions on your own, rather than verifying what is written. You would like to "own your knowledge".
- Please don't cram. This material needs time to sink in. It cannot be learned in a couple of days.
- Spend regular time on this **core course in computer science**. It will require as much time as other core courses such as CS40 and CS105 where you may spend significant time programming. In CS160, you should spend an equivalent amount of time developing a deep understanding and mastery of the core theoretical concepts and algorithm paradigms being presented.
- After doing the reading and attending class, brainstorm about homework problems, perhaps with a small study group including 2-3 other students, informally gathering ideas and without taking notes. Then remember to go off by yourself and work independently at writing up the solutions. Make sure to cite the other members of your study group.
- Use all of the course resources. Actively participate in recitation. Attend office hours when you have questions. If additional questions arise, post them to Piazza. Solve extra practice problems from the book (these may vary in difficulty). Make your own practice problems (e.g., draw an arbitrary graph and find shortest paths).
- Read all the grading feedback on your homework as well as any published solutions (e.g., recitation). The TAs are devoting many hours to grading and giving useful feedback on your work.
- Avoid memorization. Instead, focus on understanding and then being able to explain a concept or prove a theorem in your own words.
- One of the most helpful strategies is to imagine that you are teaching this material to your parent or to a cousin or to a next-door neighbor. If you don't have the details right and don't fully understand the solution yourself, you will confuse them. (Try and actually practice explaining the concepts!) This is another reason we encourage you to collaborate in small groups, to be active at recitations and to attend office hours, since these are good opportunities to discuss the course materials with other people.

Tufts University Policies

Faculty, students, and staff are jointly responsible for the policies below. Please read carefully and do not hesitate to contact us if you have further questions.

- Academic Misconduct Policy
- The Tufts Non-Discrimination Policy

Equal access

If you have a disability that requires reasonable accommodations, please contact the Office of Accessibility Services at The StAAR Center.
(Note: Academic tutoring is also offered by the StAAR Center for those who need tutoring through Tutor Finder)

Please be aware that accommodations cannot be enacted retroactively, making timeliness a critical aspect for their provision.

Academic Integrity

- It is **not** acceptable to copy solutions from **any** source for **any** work submitted in this course.
- If you cheat on an exam by consulting any source other than the allowed page of notes, the most likely penalty is a grade of F in the course.
- On a homework or project assignment, students may consult a variety of sources at the outset and groups of people may work together, discussing and strategizing how to solve the problems, **subject to the following conditions**: each person must write and submit their solutions in their **OWN** words; **every person and/or text and/or website** consulted in the process of completing the assignment must be **accurately cited** on the homework paper submitted.
- If, for whatever reason, your homework solutions end up matching solutions from any other source in a non-coincidental way, this constitutes cheating and will incur consequences.
- In case of doubt, we encourage you to reach out to the instructor.

Have Questions?

This website is a great source of information for this class. Before inquiring, please check for the answer here (e.g., when is the exam, how do tokens work, etc).
If you still have questions, we can be reached as follows:

- **In class:** Just come and chat with the instructor before/after the lectures.
- **Office hours:** We have a large pool of TAs to cover most work hours. Please keep in mind that the further away you are from a deadline, the less waiting time you will have.
- **Piazza:** The main advantage of posting your question on Piazza is that the TAs and/or your classmates will also see that question and can answer it much faster. Make sure to use the search button before posting because your question might have already been asked! If the question needs to be private, please post to "Instructors" (this will reach Karen and all the TAs).
- **PLEASE NOTE:** Your communication with TAs should be at office hours and over Piazza. **Please do NOT email TAs directly** -- their work for the department as TAs should be conducted at office hours and over Piazza, keeping their personal email addresses for their own use as students.
- **E-mail:** You can contact the course instructor using the syntax first.last@tufts.edu. However, other methods may be better than emailing in order to get the fastest response.

CS 175 Graphics (2024 Fall)

Course Number	CS 175
Semester	Fall, 2024
Hours	TR 1:30-2:45
Schedule	H+ Block
Location	JCC 140
Instructor	Remco Chang
email	remco at cs tufts edu
Office	JCC 374
Office Hours	Monday at 3pm
Graduate TA	Darby Huye
email	darby.huye at cs tufts edu
Office	JCC 440H
Office Hours	Mon: 12:00 - 1:00
Zoom	Zoom Link

Course Description

This course explores the fundamentals of computer graphics, including 3D rendering via ray casting, ray tracing, viewing transformations, 3D shape representation, and an introduction to modeling and computer animation. Assignments and projects require a good working knowledge of linear algebra and the C and C++ programming languages.

Prerequisite: CS 40 (Machine Structure and Assembly-Language Programming). Calc 2. Background in Linear Algebra a plus

Piazza Link: https://piazza.com/class/m0cnhiyn7gp2g1/

Schedule

Date	Topic	Assignments		Notes
09-03-2024	Intro			
09-05-2024	Lab 0 -- Compiling OpenGL			
09-10-2024	OpenGL			
09-12-2024	Lab1 -- Loading a Shape File			
09-17-2024	Linear Algebra Recap	A1 out		
09-19-2024	Lab2 -- Silhouette			
09-24-2024	Transform			
09-26-2024	Transform 2			
10-01-2024	Camera	A2 out	A1 due	
10-03-2024	Lab3 -- Solar System			
10-08-2024	Scene Graph | Animation			Last day to drop classes
10-10-2024	QA Session: Transformations			
10-15-2024	NO CLASS			Remco Away at VIS
10-17-2024	NO CLASS		A2 Due	Remco Away at VIS
10-22-2024	Ray Casting	A3 out		
10-24-2024	Lab4 -- Painting an Object			
10-29-2024	Illumination and Intersection Normals			
10-31-2024	Lab5 -- Dragging			
11-05-2024	NO CLASS		A3 due	Presidential Election Day
11-07-2024	Recursive Ray Tracer - Basics and Texture	A4 out		
11-12-2024	NO CLASS			Monday Schedule
11-14-2024	Shaders			
11-19-2024	Lab 6 -- Shaders: Modeling	A5 out		
11-21-2024	Final Project Discussion		A4 due (Wednesday at midnight)	
11-26-2024	NO CLASS			Remco at NYU Abu Dhabi
11-28-2024	NO CLASS			Thanksgiving
12-03-2024	Shaders II			
12-05-2024	Lab 8 -- Normal Mapping		A5 due (Thursday at midnight. Grading on Friday)	
12-10-2024	Grading A5 and Lab 8			Last day of class
12-17-2024	Final Project Presentation -- 12:00pm - 2:00pm		Final Project Due	

Textbooks

Recommended Book

R1 Fundamentals of Computer Graphics by Shirley and Marschner

Grading

Assignment 1	11%
Assignment 2	11%
Assignment 3	11%
Assignment 4	11%
Assignment 5	11%
Final Project	21%
In Class Labs (8 Labs)	24%
Total	100%

Assignments
Grading: Each assignment is worth 11% of your final grade. Out of the 11%, 2% is for your written algorithm, and 9% for the implementation.

Late Policy: All the assignments due at 11:59pm on Monday (the night before the Tuesday lecture). Exceptions will be noted on the assignment handout. The algorithm worksheets are due at noon on Fridays of the week when the assignment is handed out. Assignments that are turned in late will receive a 0. The rationale for strict deadline is that the assignments are built on top of each other (e.g. you cannot complete assignment 4 without completing assignment 3). So completing each assignment in time is essential to the success of the next assignment. If you have an extraordinary circumstance, you must contact the instructor or the TA as soon as possible and obtain written approval.

In-Class Labs
Grading: There are 8 in-class labs, each is worth 3% of your final grade. Note that Lab0 is not for credit, but getting OpenGL to compile is essential for the rest of your course work.

Late Policy: All in-class labs must be completed by the start of the next in-class lab (usually 1 week). The lab will be checked in person by the TA or the instructor during class. There is no late policy for in-class labs. You will not receive partial credit for turning in a late lab.

Final Project
Grading: Your final project will be graded in-person during the final exam period.

Late Policy: There is no late policy for the final project. You will receive a 0 if your final project isn't working by the demo day.

Accommodation

Tufts is committed to providing support services and reasonable accommodations to all students with documented disabilities. To request an accommodation, you must register with the Student Accessibility Services at the beginning of the semester. .

Acknowledgement

Some images and slides are based on lectures by Professor Andy van Dam at Brown University and Professor Daniel Keefe at the University of Minnesota.

Syllabus

CS 136: Statistical Pattern Recognition

CS 136: Statistical Pattern Recognition (SPR)
Department of Computer Science, Tufts University
Course Website: https://www.cs.tufts.edu/cs/136/2023s/
Class Meetings for Spring 2023: Mon and Wed 10:30 - 11:45am ET in JCC 180
Instructor: Mike Hughes, Assistant Professor of Computer Science
Grad TA: Xiaohui Chen, Ph.D. student, Tufts CS
Have questions? Need help?

- *Get rapid online help via our Piazza discussion forum*
- *Get in-person help at regular Office Hours*
- *For personal issues, email the instructor: mhughes(AT)cs.tufts.edu*

Quick Links: [Prereqs] [Wait List] [In Class] [Homeworks] [Collaboration Policy] [Late Work Policy] [Grading Rubric]

Course Overview

This course provides the theoretical and computational foundations for **probabilistic** machine learning. The focus is on probabilistic models, which are especially useful for any application where observed data could be noisy, sometimes missing, or not available in large quantities. We emphasize representing uncertainty with formal distributions and trying to average over these distributions when making decisions (as done in the Bayesian approach).

We will study the following probabilistic models:

- models for discrete data (e.g. next word prediction)
- linear models for regression
- generalized linear models for classification
- mixture models
- hidden Markov models for sequential data

We will try to understand all models through a unifying conceptual framework of directed graphical models.

Algorithms studied include:

- sampling via Markov chain Monte Carlo methods
- optimization via gradient descent (first-order and second-order)
- optimization via coordinate descent, as in expectation maximization and variational inference

Objectives

After completing this course, students will be able to:

- Demonstrate formal mathematical understanding of probabilistic models.
- Given an applied data analysis task, select a relevant probabilistic model, fit the model on a relevant dataset using an appropriately chosen approximate inference method, and analyze the results.
- Analyze numerical accuracy and stability of common probabilistic ML algorithms (e.g. avoiding overflow/underflow)
- Analyze scalability considerations of common probabilistic ML methods (including runtime and memory complexity requirements).

Prerequisites

This course intends to provide students a solid foundation in statistical machine learning methods.

To achieve this objective, we expect students to be familiar with the following before taking the course:

- Probability theory
-
    - e.g. you could explain the difference between a probability density function (PDF) and a cumulative distribution function (CDF)
- Basic linear algebra (comfort with matrix/vector products; have seen inverses & determinants before)
-
    - e.g. you could write the closed-form solution of least squares linear regression using basic matrix operations (multiply, inverse)
- First-order gradient-based optimization

- e.g. you could code up a simple gradient descent procedure in Python to find the minimum of functions like f(x) = x^2
- Basic supervised machine learning methods
    - e.g. you can describe the mathematical optimization problem that we solve to train linear regression and logistic regression
- Coding in Python with modern open-source data science libraries
    - Basic array operations in numpy (computing inner products, inverting matrices, etc.)
    - Making basic plots or grids of plots in matplotlib
    - Training basic classifiers (like *LogisticRegression*) in scikit-learn

Practically, this means having successfully completed at least one and ideally both of these courses (or their equivalent outside of Tufts):

- EE 104 (Probabilistic Systems Analysis)
- CS 135 (Introduction to Machine Learning)

With instructor permission, *diligent* students who are lacking in a few of these areas of coursework could be able to catch-up on core concepts via self study and thus still be able to complete the course effectively. Please see the community-sourced Resources Page for a list of potentially useful resources for self-study.

Enrolling and Wait Lists

As of the start of semester, we expect to have about 40 students enrolled in the course. We are currently at capacity, but some students may drop the course and leave openings for others (usually we see 5-10 openings in the first week of classes as schedules shift).

Our top priority is to provide each enrolled student with our full support, including the ability to get prompt answers to questions on Piazza and in office hours as well as the ability to get high-quality feedback on submitted homeworks, exams, and projects in a timely manner.

We understand some students are on the wait list (either formally on the wait list on SIS system, or just conceptually would like to be in the course). It is possible that students currently on the wait list may be added, but *only if* there is adequate staff support.

Prof. Mike Hughes will make the final decision about all wait list candidates by end of day Mon 1/30, before the ADD deadline of 2/1.

To be considered for enrollment, you must do these two things:

- Complete and submit HW0 by end of day Thu 1/26.
    - This action shows you have the necessary skills and would take the course seriously
- Message the instructor by end of day Thu 1/26 via email with subject "CS 136 Wait List Request"
    - Explain your current state within your degree program (e.g. sophomore undergraduate in CS, Ph.D. student in Math)
    - Explain why taking the course *this* semester would be important to you.

Class Format for Spring 2023

The course is organized into 5 topical units (each 2-3 weeks long), which will govern in-class and out-of-class work.

Synchronous course meetings will be **in-person** throughout Spring 2023.

As of end of first day of class (Wed 1/30) we will expect students to be signed up on Piazza. This is how we will communicate any changes to the course meeting locations with you. If you have not heard otherwise, expect course meetings and office hours to be in person in their usual locations.

What will we do in class

Each synchronous class session will occur at the scheduled time (Mon and Wed from 1030-1145am ET).

Before each class, you are expected to complete on your own time all "Do Before Class" activities posted on that day's Schedule. These will include:

- Watching video lectures (broken into 5-20 minute segments about a coherent topic)
- Complete supplementary reading from the textbook or primary research literature
- Download any relevant jupyter notebooks to prepare for in-class exercises

In class, we will typically have the following structure:

- First 5 min.: **Announcements**: Instructor updates on course logistics.
- Next 20 min.: **Recap mini-lecture**: Instructor recaps main ideas of the video lectures.
- Next 30 min.: **Small group breakouts**: Interactive exercises to solidify concepts.
- Last 20 min.: **Flex time / Quiz**: Instructor/TA available for on-demand questions. Exercises can continue if desired.

For the **small-group breakouts**, we will strive to create an exciting, interactive classroom, with lots of opportunities for students to ask questions and get feedback from the professor, TAs, and peers. Each student is responsible for shaping this environment: Please participate actively and respectfully! Show up regularly and get engaged.

Flex time is set aside specifically to acknowledge that in our flipped classroom model, you spend more time than usual outside of class preparing via readings and lectures. We will not require attendance or deliver new content in this time slot, but instead offer "office hours" like on-demand support at a time that is guaranteed available in most students' schedules.

Quiz time: Once at the end of each topical unit, we will have a *short, in-class quiz* instead of flex time. This quiz will verify your understanding of unit concepts (vast majority of questions will be math or conceptual questions, occasionally there might be a programming question). These quizzes are designed to be completable within about 15 minutes by adequately-prepared students.

What will we do outside of class?

Each of the 5 units will have the following assigned work outside of class:

- 1 written homework (HW), to build math skills (derivation and analysis, resulting in a LaTeX typeset report)
- 1 coding practical (CP), to build implementation skills (auto-graded Python exercises + a short report of figures and analysis)

Essentially, every Thursday there will be either a HW or a CP due (see Schedule).

For a complete list of graded assigned work, see the Assignments page.

Note that there is a special "first homework" (HW0) designed to make sure you have necessary prerequisite knowledge and that you become familiar with LaTeX for math report preparation.

Late work Policy

We want students to develop the skills of planning ahead and delivering work on time. To facilitate learning, we also want to be able to release solutions quickly and discuss recent assignments soon after deadlines. On the other hand, we know that this semester offers particular challenges, and we wish to be flexible and accommodating within reason.

With these goals in mind, we have the following policy:

Each student will have 240 total late hours (= 10 late days) to use throughout the semester across all homeworks and coding practicals.

For each individual assignment, you can submit beyond the posted deadline at most 96 hours (4 days) and still receive full credit. Thus, for one assignment due at Thu 11:59pm ET, you could submit by the following Mon at 11:59pm ET.

This late work deadline is key to our classroom goals. It allows us to always release homework solutions on Tuesday mornings several days before the required quiz on that unit is due, and lets us discuss the assignment in class soon after without issue.

The timestamp recorded on Gradescope will be *official*. Late time is rounded up to the nearest hour. For example, if the assignment is due at 3pm and you turn it in at 3:05pm, you have used one whole hour.

Beyond your allowance of 10 late days, zero credit will be awarded except in cases of truly unforeseen exceptional circumstances (e.g. family emergency, medical emergency). Students with exceptional circumstances should contact the instructor to make other arrangements as soon as possible.

Attendance

Participation in class is *strongly* encouraged, as you will get hands-on practice with material and have a chance to ask questions of the instructor and TAs, as well as your peers.

We only require attendance on designated **quiz** dates. Otherwise, we do not strictly require attendance at any class or track attendance.

Instructional material (readings, notes, and videos) will always be "prerecorded" and released on the Schedule page in advance, under "Do Before Class".

We will record video and audio for the main track of each interactive class session to capture important announcements and highlight key takeaways. We will strive to release that video within 24 hours on the Piazza resources page.

However, the most valuable learning interactions will occur in interactive breakout exercises, and those by design require synchronous participation.

Participation does factor in to a student's grade in two ways:

- regular in-class engagements (being a memorable participant in 8 or more breakout sessions throughout the semester)
- regular Piazza forum discussions (posting 8 or more times throughout the semester)

Exams

In Spring 2023, beyond the 5 unit-specific quizzes, there will be no required exams.

However, we will offer one **optional** exam (on the scheduled final exam date) as a way to allow students who missed or under-performed on a quiz for a particular unit a chance to improve their scores. Essentially, taking this exam can only improve your score.

Project

In the last month of the course, there will be an open-ended small project that allows you to apply and synthesize your knowledge about probabilistic modeling.

Students will work in pairs, and will gain practical experience in selecting, applying, evaluating and improving the models we study in this class in the context of a real dataset.

Workload

Each week, you should expect to spend about 8-12 hours on this class.

Here's our recommended break-down of how you'll spend time each week:

- 1.75 hr / wk preparation before Mon class (reading, lecture videos)
- 1.00 hr / wk active participation in Mon class (excludes flex time)
- 1.75 hr / wk preparation before Wed class (reading, lecture videos)
- 1.00 hr / wk active participation in Wed class (excludes flex time)
- 4.00 hr / wk on HW or CP assignment (one assignment due each week, takes ~4 hr total)
- 0.50 hr / wk preparing for and taking quiz (quizzes happen every few weeks, so get 0.5 hr prep plus 0.5 time taking the quiz)

This totals to ~10.00 hr / wk.

Grading

Final grades will be computed based on a numerical score via the following weighted average:

- 25% homeworks (3% HW0, 4.4% each for HW1-HW5)
- 22% coding practicals (4.4% each for CP1-CP5)
- 36% short quizzes (one per unit, averaged after dropping the lowest quiz)
- 15% project (including final report and intermediate checkpoint reports)
- 2% participation in class and on Piazza discussion forum

When assigning grades given a final numerical score (from 0.0 to 1.0), the following scale will be used:

- 0.94-1.00 : A
- 0.90-0.94 : A-
- 0.87-0.90 : B+
- 0.83-0.87 : B
- 0.80-0.83 : B-
- 0.77-0.80 : C+
- 0.73-0.77 : C
- 0.70-0.73 : C-
- 0.67-0.70 : D
- 0.63-0.67 : D
- 0.60-0.63 : D-
- Below 0.60: F

The highest possible grade of "A+" will be awarded at the instructor's discretion.

We do not *round up* grades. This means you must earn at least an 0.83 (not 0.825 or 0.8295 or 0.8299) to earn a B.

Textbook

As a primary textbook, we will use "Pattern Recognition and Machine Learning" by Christopher M. Bishop.

A free PDF is available online from the author:

<https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf>

Other suggested resources can be found on the Resources Page.

Collaboration Policy

Our ultimate goal is for each student to fully understand the course material. With this goal in mind, we have the following policy:

For quizzes, all work should be done individually, with no collaboration with others whatsoever.

For homeworks and coding practicals (HWs and CPs), we have the following policy for student work:

> *You must write the text of anything that will be turned in -- all code and all written solutions --* ***on your own*** *without help from other human or artificial agents. You may not share any code or solutions with others, regardless of if they are enrolled in the class or not.*
>
> *We do encourage high-level interaction with your classmates. After you have spent at least 10 minutes thinking about the problem on your own, you may verbally discuss assignments with others in the class. You may work out solutions together on whiteboards, laptops, or other media, but you are not allowed to take away any written or electronic information from joint work sessions with others. No notes, no diagrams, and no code. Emails, text messages, and other forms of virtual communication also constitute "notes" and should not be used preparing solutions.*
>
> *When preparing your solutions, you may always consult textbooks, materials on the course website, or existing content on the web for general background knowledge. However, you cannot ask for answers through any question answering websites such as (but not limited to) Quora, StackOverflow, ChatGPT, etc. If you see any material having the same problem and providing a solution, you cannot check or copy the solution provided. If general-purpose material was helpful to you, please cite it in your solution within your "Collaboration Statement".*

Collaboration Statement

At the top of every turned in homework (HW) and coding practical (CP), you must include the names of *any* people you worked with, and in what way you worked them (discussed ideas, debugged math, team coding). We may occasionally check in with groups to ascertain that everyone in the group was participating in accordance with this policy.

Academic Integrity Policy

This course will strictly follow the Academic Integrity Policy of Tufts University. Students are expected to finish course work independently when instructed, and to acknowledge all collaborators appropriately when group work is allowed. Submitted work should truthfully represent the time and effort applied.

Please refer to the Academic Integrity Policy at the following URL: https://students.tufts.edu/student-affairs/student-life-policies/academic-integrity-policy

Accessibility

Tufts and the instruction team of CS 136 strive to create a learning environment that is welcoming students of all backgrounds.

If you feel unwelcome for any reason, please talk to your instructor so we can work to make things better. If you feel uncomfortable talking to members of the teaching staff, consider reaching out to your academic advisor, the department chair, or your dean.

Please see the detailed accessibility policy at the following URL: <https://students.tufts.edu/student-accessibility-services>

DOWNLOAD SYLLABUS

Download the most complete and up-to-date PDF version for the **in-person course here**. For the **online course** download **here**

COURSE GOAL

By the end of the semester, students should be able to:

1. Identify the major classical and modern AI paradigms and explain how they relate to each other
2. Analyze the structure of a given problem such that they can choose an appropriate paradigm in which to frame that problem
3. Implement a wide variety of both classical and modern AI algorithms

CS131 Artificial Intelligence

HOMEWORK & FINALS

ASSIGNMENTS **Six (6)** assignments (roughly matching the major course sections) will be given during the class.

Python is the official implementation language for assignments. C++ or other languages can be used ONLY after negotiating with the teaching staff before submitting.

You can submit a ZIP file with all the files needed to run your solution on Canvas. There is no auto-grader. All assignments are manually executed and graded. For Python, provide a plain PY file (**DO NOT** use Jupyter notebooks). If you are writing in C++, please include a CMakeLists.txt file and other compilation instructions.

Your solutions may make use of any numerical libraries for pre-processing and visualization. However, the core portion of your solutions **MUST** be implemented from scratch. Any material regarding the solution will be submitted electronically. Homework is due at midnight of the deadline (check the **COMP 131 website calendar** for more details). Late assignments are penalized at 10% for each 24-hour delay. No homework will be accepted after one week.

You have **three (3)** 4-day extensions available to you. Multiple extensions cannot be combined on a single assignment. If you decide to use them, please alert the teaching staff **before the assignment's deadline**.

HOMEWORK Students will be asked to read a section from the textbook for reading assignments. Occasionally, an exercise or two will be given in class to be solved at home. The results of the homework will be discussed upon request from the students.

ATTENDANCE **(Only for the in-person class)** For some lectures, I will provide a short questionnaire that must be returned at the end to promote engagement and obtain feedback on the class's understanding of the topics. The questionnaire is based on best effort, so there is no actual grade. However, it will impact the attendance portion of your grade. You can miss at most two questionnaires without any impact on the grade.

TESTS **three (3)** tests will be administered over the semester (see the tentative schedule below). The tests will combine single-choice, multiple-choice, calculations, and open-ended answers. Books must be closed, and electronic devices are not allowed.

If you need to miss any of the tests for any reason, you must inform me before the scheduled day so that a make-up session can be arranged.

Please Note: If you wish to dispute a grade, you must do so within one week of receiving the grade. After such a term, the grade will be considered final.

PLEASE NOTE If you wish to dispute a grade, you must do so within one week of receiving the grade. After such a term, the grade will be considered final.

You must show proficiency in all grading areas to pass the class. A failing average (below 70 or "C-") in any of the grading areas (assignments or tests) will result in a failing grade in the class. Your final grade will be determined using the following percentage breakdown: **(in-person course)** 45% coding assignments, 50% tests, and 5% attendance; **(online course)** 50% coding assignments and 50% tests.

Please, see PDF version of the syllabus for further details regarding standard grading scale. No grading on the curve will be applied.

ACADEMIC HONESTY

Science is, at its core, a collaborative effort. The advantages of coming together for examination or comparison, sometimes even to explain the problem, are well known. I strongly encourage students to discuss course material, problems, and applications outside the classroom with the teaching staff and other students. You are also encouraged to form study groups for the tests.

Integrity and honesty, however, are equally important qualities of any future academic, scientist, or engineer. We take plagiarism very seriously. You must do your homework and the final projects independently and without the help of any AI-enabled tools. If you need help, the teaching staff will be more than happy to help you!

The Faculty of the School of Arts and Sciences and the School of Engineering must report suspected academic integrity violations to the Dean of Student Affairs Office. If I suspect you cheated or plagiarized in this class, I must report the situation to the dean. If students do not understand these terms or those outlined in the Academic Code of Conduct, they must talk to the instructor.

STUDENTS WITH DISABILITIES

To maximize each student's participation in the Tuft experience, Tufts and the teaching staff are committed to providing equal access and support to all qualified students through reasonable accommodations.

If you need reasonable accommodations for a disability, please contact the Student Accessibility Services office at **ACCESSIBILITY@TUFTS.EDU** or **(617) 627-4539** to determine appropriate actions.

Please be aware that accommodations cannot be enacted retroactively, making timeliness a critical aspect for this provision. If you need special accommodations for exams, please do not wait until just before the exam to contact SAS. Please, give sufficient time to arrange the necessary accommodations.

FEEDBACK

Tufts and the Artificial Intelligence teaching staff strive to create a learning environment that is welcoming to students of all backgrounds.

Your thoughts and concerns are important. You are encouraged to give feedback to the instructor throughout the term. If you feel uncomfortable talking to members of the teaching staff, consider reaching out to your academic advisor, department chair, or dean.

Tufts CS 117:
Internet-scale Distributed Systems

Fall 2025

For Fall term 2025, admission to this class is by permission of the instructor only; students cannot sign up in SIS without prior authorization. ***To be considered for admission, you must fill out an online application form by Friday April 4th, which is several days before SIS opens for registration.*** **The course is usually over-subscribed, and students will be selected based on the information provided in the online application; you will not be able to register in SIS unless the Professor has specifically authorized your student ID.**

The official deadline for applications has passed. On-time applications received by end-of-day April 4th are being processed now. ***A limited number of seats will be held until after juniors register in SIS, so that we may consider applications from students who did not hear about the application process. If you want to take CS 117 and have not already applied, please submit your application immediately!***

A Web page is available with more detailed instructions for applying. Note that due to the designation of CS 117 as a high demand course, SIS may show available seats even when the course is already full. Students who apply using the online will receive regular updates via email on the admissions process.

Are you trying to decide whether CS 117 is right for you? If so, please visit the information page for prospective CS 117 students. It gives information about the course content, pre-requisistes, workload, etc.

Course description

The World Wide Web, one of the most important developments of our time, is a unique and in many ways innovative distributed system. This course will explore the design decisions that enabled the Web's success, and from those will derive important and sometimes surprising principles for the design of other modern distributed systems.

We will introduce and draw comparisons with more traditional distributed system designs, including distributed objects, client/server, pub/sub, reliable queuing, etc. We will also study a few (easily understood) research papers and some of the core specifications of the Web. Specific topics to be covered include: global uniform naming; location-independence; layering and leaky abstractions; end-to-end arguments and decentralized innovation; Metcalfe's law and network effects; extensibility and evolution of distributed systems; declarative vs. procedural languages; Postel's law; caching; and HTML/XML/JSON document-based computing vs. RPC.

The purpose of this course is not to teach Web site development, but rather to explore lessons in system design that can be applied to any complex software system.

Logistics

Classes will meet Tuesdays and Thursdays from 4:30 PM - 5:45 PM US Eastern time, in room JCC 160.

Students are expected to attend all classes in person, and participation in class discussions is a component of your grade. Video recordings of individual classes will be made available *only* to students who have excused reasons for absence (illness, job interview, etc.) Please contact the Professor, in advance when practical, to have your absence approved and to get access to the corresponding recording. Classroom sessions will not be live streamed.

Instructor, TAs and Getting Help

This course will be taught by Professor Noah Mendelsohn. Noah's office hours will be listed on his home page. Our teaching assistant(s) for the course is/are Hamza Khalid, Owen Thomas, and Charlotte Versavel. Email sent to ta117@cs.tufts.edu will reach all of us. When you do need to contact Noah Mendelsohn only, you can e-mail him at noah@cs.tufts.edu.

For course news and Q/A we will be using Piazza (click here to enroll). If you have questions, please post them there.

Books

There are no books that you are required to buy, but several you might want to consider. Your alternative is to use Safari Books Online, an online eBook service to which Tufts students have access. Information about books and using Safari is available on the CS 117 Information Page.

Collaboration, use of Others' Work or AI Tools, and Academic Misconduct

Students should read the policy on Academic Integrity available at http://students.tufts.edu/student-affairs/student-life-policies/academic-integrity-policy. Absolute adherence to the Code of Conduct is demanded of the instructors, teaching assistants, and students. This means that no matter the circumstance any misconduct will be reported to Tufts University. You are especially warned that code is not to be shared, except with your partner(s), and *all* quotations in your written work must be properly set off with quotation marks and appropriate bibliography citations.

You are strongly encouraged to discuss problems and to brainstorm with friends and colleagues. This course is about broad architectural themes in computing systems. You will learn them much better if you discuss and debate them with other students, and with the course staff. However, *except in cases where an assignment specifically allows you to collaborate with others or to adapt existing code,* the code and other written work you submit must be entirely your own.

You must only submit writing and code that is your own original work, or jointly the work of you and your partner in the case of pair-programming projects. *You must not submit writing or code that was created by or adapted from the output of ChatGPT or similar "artificial intelligence" tools*. You must ensure to the extent that you reasonably can that sources you reference are not the result of such tools, unless those results have been curated and fact-checked by responsible and knowledgeable humans. If your work is in any way suggested by or influenced by such AI tools, then you must clearly explain when making your submission. Failure to follow these rules is typically a violation of the Tufts rules on Academic Integrity, and will be reported to the appropriate authorities.

In all cases where you use or adapt the materials of others, you must respect applicable copyrights. In general, such copyrights do not prevent you from using small fragments of a line or two of code or text, but for anything larger, you must make sure that the license accompanying the code or written work allows you to copy it.

Every source of assistance must be acknowledged in writing. There is never a penalty for seeking help with a problem, but help must be acknowledged, and you must only submit work that you completely understand. You will be graded on what you have learned and on the quality of your own contributions. If you have any doubt as to whether what you're doing is appropriate *it is your responsibility to check in advance!*

Programs and written material you prepare for this course must not be posted online, except when the assignment specifically requires that you do. For example, it is not acceptable to post your assignment solutions for public access on GitHub. You are encouraged to submit general programming questions to online forums such as [Stack Overflow](). Questions about particular homework problems must never be posted to public fora — send an e-mail to the course staff.

(This policy is adapted from the one used by Ming Chow for the summer 2012 edition of COMP-15.)

CS 116: Introduction to Security

Tufts University Department of Computer Science, Spring 2025

Course Description

A holistic and broad perspective on cyber security. Attacking and defending networks, cryptography, vulnerabilities, reverse engineering, web security, static and dynamic analysis, malware, forensics. Principles illustrated through hands-on labs and projects, including Capture The Flag (CTF) games.

Sections

1. In-person, undergrads and grads: Tuesdays, 4:30 - 5:45 PM EST in Cummings Center Room 270; [Thursdays on Twitch](), 4:30 - 5:45 PM EST
2. Online Master's in Computer Science: live sessions on Wednesdays, 5:30 - 7:00 PM EST (via Zoom)

Instructor

- Ming Chow, ming.chow@tufts.edu
- Office Hours: Tuesdays, 3:00 - 4:15 PM EST in JCC, fourth floor --by the kitchen area

Prerequisites

- CS 15 or Data Structures equivalent course. Recommended (not required) that you have taken CS 30 or 40. Please disregard prerequisite that CS 40 is required as listed in the University's bulletin as they are incorrect!

Hardware and Software for This Class (on your personal computer)

Absolute Requirements

- A modern web browser (e.g., Firefox, Google Chrome, Chromium, Safari, Microsoft Edge)

- A command line interface to run Unix/Linux commands (e.g., macOS, Windows with Linux Subsystem, a Linux-based virtual machine, a Docker container)

List of Security Tools That Will Be Used in Course

The following is a list of security tools that will be used in the course. All of these tools are platform-independent.

- Wireshark
- Nmap
- Netcat
- Python
- Scapy
- John the Ripper
- Burp Suite
- apktool and Java as apktool requires Java

Assessment

- Labs (80%)
- Quizzes (20%; there will be two)

Course Infrastructure

- Lab submissions: Canvas (Tufts UTLN required)
- Quizzes: Canvas (Tufts UTLN required)
- Announcements: Piazza
- Discussions: Piazza

Syllabus

Important note: always follow Canvas for due date on labs!

Topic 1, starting week of January 15th	• Course Introduction - By the end of this week, students will learn many of the fundamental Linux commands, an important skill for any good security practitioner, by playing	• Lab 1: Working with the Command Line ◦ Publicly accessible version of Lab 1

	Capture The Flags via OverTheWire. Students will remember the three principles of the CIA triad, critical to any organization's security infrastructure. • Readings and Videos • Video on YouTube with Closed Captioning: The Command Line Interface	

Topic 2, starting Tuesday, January 21st	• Networking and Packets - By the end of this week, students will be able to dissect packet captures (PCAPs), network traffic.
	• Readings and Videos
	• Thursday, January 23rd: Packet Analysis Using Wireshark
	  ◦ On Twitch
	  ◦ On YouTube with Closed Captioning
		• Lab 2: Packet Sleuth
		  ◦ Publicly accessible version of Lab 2

Topic 3, starting Tuesday, January 28th	• Attacking Networks - By the end of this week, students will perform network reconnaissance and port scanning, and build a rudimentary Security Information and Event Management (SIEM) / intrusion detection system (IDS).
	• Readings and Videos
		• Lab 3: Scanning and Reconnaissance
		  ◦ Publicly accessible version of Lab 3
		• Lab 4: Snake Oil, The Incident Alarm
		  ◦ Publicly accessible version of Lab 4

	• Thursday, January 30th: Reconnaissance using Ping, Netcat, and Nmap
	    ◦ On Twitch
	    ◦ On YouTube with Closed Captioning
	• Thursday, February 6th: Distributed Denial of Service (DDoS) Attacks and Scapy feat John Hammond
	    ◦ On Twitch
	    ◦ On YouTube with Closed Captioning	

Topic 4, starting Tuesday, February 11th	• Cryptography - By the end of this week, students will be able to crack passwords on a Linux or Windows system, use one-way hash functions, and briefly describe how Transport Layer Security works.
	• Readings and Videos
	• Thursday, February 13th: Passwords and Password Cracking with John the Ripper
	    ◦ On Twitch
	    ◦ On YouTube with Closed Captioning
		• Lab 5: Password Cracking Contest opens, due on Friday, March 7th at 11:59 PM PST

Topic 5, starting Tuesday, February 18th	• Vulnerabilities - By the end of this week, students will know the difference between CVE and CWE. • Readings and Videos	• Quiz 1 open on Monday, February 17th; due Sunday, February 23rd at 11:59 PDT

Topic 6, starting Tuesday, February 25th	- Web Security - By the end of this week, students will able to perform and defend against the following attacks: Cross-Site Scripting (XSS), SQL injection, Cross-Site Request Forgery (CSRF), session hijacking, cookie tampering, directory traversal, command injection, remote and local file inclusion. - Readings and Videos - Thursday, February 27th: SQL Injection and Web Proxies - On Twitch - On YouTube with Closed Captioning - Thursday, March 6th: Vulnerability Scanning, Exploitation, Badness-O-Meter - On Twitch - On YouTube with Closed Captioning	- Lab 6: The XSS Game - Publicly accessible version of Lab 6 - Lab 7: Gain Access to Website - Publicly accessible version of Lab 7
Topic 7, starting Monday, March 24th	- The Capture The Flags (CTF) Game Played Online - By the end of this week, students will be able to find and take advantage of a number of vulnerabilities on a live web application. - Readings and Videos	- Lab 8: The CTF Write Up - Publicly accessible version of Lab 8

Topic 8, starting Tuesday, April 1st	• Static and Dynamic Analysis - By the end of this week, students will be able to perform static analysis and dynamic analysis scans on software, write a technical risk analysis that is communicated to upper management.
	• Readings and Videos
	• Thursday, April 3rd: Really, Really Bad Code and Static Analysis
	  ◦ On Twitch
	  ◦ On YouTube with Closed Captioning
		• Lab 9: Technical Risk Analysis

Topic 9, starting Tuesday, April 8th	• Malware - By the end of this week, students will be able to describe types of malware, see certain malware behaviors, scan and analyze malware, reverse engineer Android apps to determine if they are malicious.
	• Readings and Videos
	• Thursday, April 10th: Malware and Malware Analysis
	  ◦ On Twitch
	  ◦ On YouTube with Closed Captioning
		• Lab 10: Android Malware Analysis
		  ◦ Publicly accessible version of Lab 10

Topic 10, starts Tuesday, April 15th	• Forensics and Incident Handling - By the end of this week,	• Quiz 2 open on Monday, April 14th; due Tuesday, April

	students will be able to acquire data from a disk (e.g., USB drive) using **dd**, analyze image of disk from \`dd\` using forensics tools, and recover deleted files off a disk.
	• Readings and Videos
	• Thursday, April 17th: Forensics
	  ◦ On Twitch
	  ◦ On YouTube with Closed Captioning
		22nd at 11:59 PM PDT

Topic 11, starts Tuesday, April 22nd	• ~~The Future: Nihilism or Hope? - By the end of this week, students shall debate and ponder the hard questions in security, and be able to argue multiple viewpoints.~~ Lessons Not Learned: We Can't Even Get the Basics Right
	• Readings and Videos
	• Thursday, April 24th: Opportunities, Where Do You Go From Here
	  ◦ On Twitch
	  ◦ On YouTube with Closed Captioning	

Topics That Will Not Be Covered In This Course

- Social Engineering
- Privacy

Frequently Asked Questions

Q: Is there a textbook for this course?

A: No

Q: Are there teaching assistants (TAs) for this course?

A: No

Q: What is the workload of this course?

A: Here is a list of all the labs with expected length and difficulty:

- Lab 1: Working with the Command Line, Short (1 hour max) to Long (3+ hours) --you can put in as much time as you want on this lab
- Lab 2: Packet Sleuth, Medium (1 - 3 hours)
- Lab 3: Scanning and Reconnaissance, ~~Very short (30 minutes)~~ Your mileage may vary on this lab, could be 30 minutes, could be 2 hours or more. **NOTE: This lab cannot be made publicly available because an actual target is used.**
- Lab 4: Python and the Incident Alarm, Long (over 3 hours) to Impossible
- Lab 5: The Password Cracking Contest, If you crack all the password hashes (read: good luck with that), you will receive an automatic "A" in the course
- Lab 6: The XSS Game, Medium
- Lab 7: Gain Access to Website, Very short. **NOTE: This lab cannot be made publicly available because an actual target is used.**
- Lab 8: The CTF Game, One week --team based. **NOTE: This lab cannot be made publicly available because an actual target is used.**
- Lab 9: Technical Risk Analysis, Short to Medium.
- Lab 10: Android Malware Analysis, Short to Medium

Q: Does this course count towards the M.S. in Cybersecurity and Public Policy?

A: Yes

Q: Does this course count towards the M.S. in Software Systems Development?

A: Yes. In fact, this is one of the four required courses for the M.S.

Q: Did you remove information on using Kali virtual machine for this class? If so, why?

A: Yes. After all these years, it was more trouble than it was worth. Further reasons:

1. Accessibility. For students who are visually impaired, using a virtual machine can be very difficult.

2. Not all students have a capable laptop. Sometimes due to financial reasons, some students use Chromebooks. The tools required for this course can be installed natively on macOS, Windows, and Linux.
3. Performance. Sometimes, using a VM can be very slow. A VM also do not use native drivers (e.g., for networking).
4. Hard disk space requirement: at least 10 GB necessary.
5. Apple M1 Macs cannot run most Intel x86 virtual machines.

Q: Is Piazza used in this course?

Yes, quite a lot

Q: Why is there a course website and a course Canvas? If you say "it is a nuisance for students to use multiple websites and services for one course", what gives?

This course website serves a few critical purposes. Years ago, I made a decision to make all the readings, slide decks, and most of the labs publicly available. The reasons: (1) to show that Tufts is serious and is working on Cyber Security matters, (2) to provide learning material to the public on Cyber Security as the Cyber Security education problem is very dire, (3) for recordkeeping on what is taught and not taught in this Security class --this comes up often when we speak to industry and organizations who want to work with Tufts on Cyber Security-related matters. The Canvas site for this course isn't made publicly available. Even if Canvas site was made publicly available, content is behind a walled garden, and (4) for redundancy if Canvas goes down.

Q: I have not taken a course on Networks (CS 112), Operating Systems (CS 111), or Computer Architecture (CS 40) yet. Is that a problem?

No. Cyber Security is a very broad field and it is impossible for anyone, even professionals, to know everything. What is important for you is to start thinking about Security.

Q: Will videos be recorded in case I miss class?

If you miss the in-person Tuesday classes due to illness or personal matters, you can always watch the recorded Wednesday sessions for online Master's students (Wednesday sessions are always recorded). Tuesday and Wednesday sessions are practically identical. Thursdays are on Twitch which are recorded and also exported to YouTube.

Q: If I am taking this course for professional purpose, can I have a tuition reimbursement letter or certificate?

A: [Absolutely! It's a nice tuition reimbursement letter, hand signed!]

If you have read this far, send me an email (ming.chow AT tufts DOT edu) with the subject "¯_(ツ)_/¯" to earn a reward.

Course Policies

Labs

- With the exception of the password cracking lab and CTF writeup, all labs for a given topic, are due on a Sunday at 11:59 PM PDT (that is, Pacific Time).
- With the exception of password cracking lab, the CTF game, and quizzes, you are granted an automatic extension of 24 hours at no cost (i.e., grace period). A lab or quiz submitted after the grace period will not be accepted.
- No extension tokens.

Accessibility Statement

Tufts is committed to providing equal access and support to all qualified students through the provision of reasonable accommodations so that each student may fully participate in the Tufts experience. If you have a disability that requires accommodations, please contact the StAAR Center (formerly SAS) staarcenter@tufts.edu or 617-627-4539 to make an appointment to determine appropriate accommodations. Please be aware that accommodations cannot be enacted retroactively, making timeliness a critical aspect for their provision. You can learn more about the StAAR Center at https://students.tufts.edu/student-accessibility-services.

Expectations and Structure of This Course

This course will be a fun one for sure. A few notes on the expectations and structure of this course:

1. You are responsible for your own learning.

A very important point: if you want everything gone over in lecture or in notes, then this is not the course for you. More importantly, that's not how things work in real life.

2. You will learn by doing.

Each week, there will be at most three labs to hone your skills and to aim at the crux of the matter for the week. Here's an analogy: you don't learn how to cook simply by just reading cookbooks and watching YouTube videos. You learn by making, using your hands, and making mistakes.

3. You will learn by asking questions.

It is your responsibility to ask questions early and to ask for help...

4. ...and I expect discussions online to be very active and civil.

Share thoughts and respond to other people's questions. I will be online constantly. It is no secret that I respond very quickly unless I need to be away.

created by human
not by AI

CS 105: Programming Languages

Spring 2025

https://www.cs.tufts.edu/comp/105

Class Sessions	Monday, Wednesday 3:00pm–4:15pm
	JCC 270
Instructor	Richard Townsend, richard.townsend@tufts.edu
Richard's Office Hours (JCC 440A)	Tuesdays 2:00pm-4:00pm
	(or by appointment)
Final Exam	Friday, May 2, 3:30pm-5:30pm, JCC 270

Course Overview

Summary

CS 105 introduces you—through extensive practice—to ideas and techniques that are found everywhere in today's programming languages. You will learn high-level, flexible programming skills that are applicable to older languages, popular modern languages, and even languages that don't yet exist. No matter what language you work in, when you finish 105, you'll be writing more powerful programs using less code. At the same time, you will learn the mathematical foundations needed to talk precisely about languages and programs: abstract syntax, formal semantics, and type systems.

You will explore and apply programming language concepts through multiple case studies, conducted using (mostly) tiny languages that are designed to help you learn. In any given case study, you may act as a practitioner (by writing code in a language), as an implementor (by working on an interpreter for the language), as a designer (by inventing semantics for a related language), or as a scholar (by proving mathematical properties of the language).

Logistics

The main point of access for the course is the website (the link is at the top of this document): it provides all the slides, readings, homeworks, solutions, and recitation materials.

Here is the weekly flow of the course:

1. Come to class sessions: take notes, participate in partner-based activities, and ask questions! This will ensure you're prepared for assignments and recitations. **Recitations will not be useful to you if you haven't come to class first.**

2. Read over the homework spec *before* attending recitation. Ask questions about it on Piazza and in office hours (OH), or talk about it with your classmates. Start the assigned reading.

3. Go to recitation, where you will get supervised practice working on problems that resemble the homework problems.

4. Work on your assignment, first by doing the reading and comprehension questions and then by completing the programming and/or proof problems. Make sure you spread this out so you have breaks between your 105 work. Post on piazza or come to OH if you need assistance.

If you feel like you're falling behind or are overwhelmed with 105, please reach out to a member of the course staff; we want to help you!

Prerequisites

- CS 15 Data Structures: You need substantial programming experience to keep up with the homework, particularly working with dynamically allocated (i.e., heap-allocated) data structures and recursive functions. We will also be referencing many classic data structures (stacks, queues, lists, trees) and algorithms (sorts, graph algorithms) throughout the course.
- CS/MATH 61 Discrete Math: You need some experience proving theorems, especially by induction. You should also be comfortable reading and writing formal mathematical notation.

Course Goals

By the end of this course, students should be able to

1. Design code using algebraic laws.
2. Read and write code that uses functional programming techniques.
3. Recognize the merits of polymorphism, type checking, and type inference.
4. Use functional and object-oriented language features to hide implementation details.
5. Understand precise specifications of how programming languages work.
6. Mathematically prove the behavior of programs written in a given language.

Technical Resources

Office Hours

This course is challenging, but we want to help you succeed! If you need help understanding a concept or tackling an assignment, or dealing with a pernicious bug, you can participate in our TA office hours sessions. You can also attend an instructor's office hours for any of these reasons, to discuss any issues you're having with the course, or just to say "hi!"

TA office hours will take place in the middle of the main hallway on the 3rd floor of the Cummings Center (near the stairs). When you want help, write your name and your place in the queue (as a number next to your name) on the whiteboard.

Office hour assistance is meant to help you along or provide a nudge in the right direction; we expect that you will try to grapple with your issue yourself before asking for help. To help you follow this practice, you are expected to provide a specific topic or question to receive help. Here are some *poor* examples of specific topics/questions: "why doesn't my program work?", "debugging", "can you explain impcore?". Here are *much better* examples: "why does my function produce a number when it should produce a list?", "debugging a syntax error", "how do I call a function in impcore?".

The full office hours schedule is posted and kept up to date as a pinned Piazza post; make sure you check it before trekking to Cummings!

Piazza

We will be using Piazza for class discussion; you can get to the 105 page via this link or on the Home page of the course website. **Piazza will host all of our major course announcements; it is your responsibility to check in regularly to avoid missing any important information.**

In general, you should post all questions related to the course on Piazza; post publicly if at all possible, as other students may have similar questions or be able to help you faster than the course staff. Please post privately to the course staff if your question reveals individual work (e.g., algebraic laws or code) or concerns your grades.

Textbooks

This course has two required textbooks:

- *Programming Languages: Build, Prove, and Compare.* Norman Ramsey. Tufts University, Fall 2022.

  You must have this specific edition; if you're looking at an old version it may no longer align with this semester's assignments! The correct edition can be purchased from the Tufts Bookstore. If paying for this book is a financial hardship or you have any other questions about obtaining the textbook, please contact Sarah White (she's great) at s.white@tufts.edu.

- *Seven Lessons in Program Design.* Norman Ramsey.

  This short booklet is linked from our course website (at the bottom of the home page).

There is also one optional textbook that is useful for the few ML assignments we have:

- *Elements of ML Programming, ML97 Edition.* Jeffrey Ullman. Pearson, 1997.

Tentative Schedule

The tentative schedule for the course is displayed on our course website. Topics and assignments are subject to change, and may be updated as the semester progresses (you will be informed of any changes as soon as they happen). Typically: Lectures will be on Mondays and Wednesdays; homeworks will be due on Tuesdays at 11:59pm and new homeworks will go out Wednesday mornings; and recitations are either on Thursday or Friday (depending on which recitation you are enrolled in). **There are exceptions to all of these throughout the semester** (e.g., sometimes homework will be due on different days, etc.). Therefore, it is important for you to keep up with the schedule linked above.

Assessments and Grading

Your course grade will be produced as a weighted sum of your scores in four categories:

1. Recitations (5%)
2. Homework: Comprehension Questions (5%)
3. Homework: Programming and Proof Problems (70%)
4. Final Exam (20%)

All grades will be posted on Gradescope for students to review.

Although your ultimate course grade will be a conventional letter, much of your specific work will be graded on a scale of No Credit, Poor, Fair, Good, Very Good, and Excellent. These grades correspond to numeric values of 0, 65, 75, 85, 95, and 100, respectively. For example, if an assignment has 3 equally weighted components and you received a Good on two parts and a Very Good on the third, your grade for that assignment would be approximately $.33 * 85 + .33 * 85 + .33 * 95 = 87\%$. In a typical class, a consistent record of Very Good work will lead to a course grade in the A range. Work rated Good corresponds to a wide range of passing grades centered around B. Work rated Fair will lead to low but satisfactory course grades around a C; if a significant fraction of your work is Poor, you can expect an unsatisfactory grade.

Recitations

Recitations are graded as Attended or Not Attended (a recitation TA will take attendance in each session); your final recitation grade is the percentage of recitations you attended. We drop your two lowest recitation grades, effectively letting you miss two recitations at no penalty. There is no such thing as an "excused absence" from recitation, so if you need to miss one just let your recitation leader know. If you need to miss more than two recitations due to a personal emergency, please explain the situation to your academic dean and ask them to contact a course instructor.

Homework

In general, homeworks will go out every Wednesday and be due at 11:59:00PM EST the following Tuesday (there will be a few larger assignments in the second half of the course that are exceptions to this rule). Each homework will have two components:

1. Your responses to a set of reading comprehension questions (CQs), which will be submitted as a .txt file.
2. A programming and/or proof component.

You will submit all of your work directly to Gradescope. No submissions will be accepted via email or other means. To avoid missing the strict 11:59:00PM EST deadline, you should submit whatever you have a bit earlier (or much earlier!) even if incomplete; you may submit as many times as you wish before the deadline. We will always grade the latest submission.

After submitting to Gradescope, you must always look over the files you submitted to make sure that they are what you expect. In particular, make sure: (1) you submitted all necessary files; (2) the code/files you submitted are the latest versions of your work; and (3) any PDFs you submitted are readable and not corrupted. (3) is important, since VSCode has been known to corrupt PDFs that are synced with the department servers. **It is your responsibility to check that the files you submitted are the correct and readable versions. Regrade requests will not be considered for submitting incorrect files.**

Comprehension Question Grades A set of comprehension questions (CQs) is associated with each homework assignment. Your answers to these questions are graded based on the number of questions answered and the correctness of each answer, roughly following this scale:

- **Excellent**: All answers are completely correct.
- **Very good**: All answers are mostly correct.
- **Good**: A majority of answers are mostly correct.
- **Fair**: Most answers are incorrect, but all were reasonably attempted.
- **Poor**: Most answers are incorrect or blank.
- **No Credit**: No answers are provided.

Programming and Proof Problem Grades The programming and proof problems on each assignment will be graded both for correctness (e.g., Do your programs pass our automated tests? Are your theoretical proofs and rules accurate and complete?) and for structure and organization (e.g., Does your code follow our course coding standards? Are your algebraic laws accurate and complete?). Correctness is typically evaluated by automated testing scripts that list which tests you passed or failed and why. Structure and organization is evaluated manually by our course staff. Both of these components are graded with a coarse five-point scale:

- **Excellent**: Your submission is outstanding in all respects.
- **Very Good**: Your submission does everything asked for, and does it well. There may be something small that could be improved.
- **Good**: Your submission demonstrates quality and significant learning.
- **Fair**: Your submission is quite lacking in one or more aspects; key issues need to be addressed. This is the lowest satisfactory grade.

- **Poor**: Your submission shows little evidence of effort or has other serious deficiencies. This is an unsatisfactory grade.
- **No Credit**: No submission provided OR the submission exhibits plagiarism OR the submission violates a rule that the homework indicated would result in no credit.

Late Policy

For every assignment we have, you are allowed to submit up to 24 hours after the posted deadline. There is no penalty for submitting within this 24-hour grace period. However, typically a new assignment will be released immediately after each deadline—so by submitting late, you lose some time on the next assignment. If you submit more than 24 hours after a deadline, your submission will not be accepted.

You should think of these late submissions as extensions you have been granted ahead of time and use them when you might have otherwise tried to ask for an extension. There are no additional "late tokens" in this course. You can think of our late policy as granting one "late token" per assignment.

If you experience an extraordinary difficulty, such as serious illness, family emergencies, or other extraordinary unpleasant events, your first step should be to contact your advising dean as soon as you can: explain the situation to them and ask them to contact a course instructor. Your dean will work with the instructor to make appropriate arrangements. **IMPORTANT: The earlier you notify your dean, the more flexibility the course staff will have to make appropriate arrangements.**

Regrades and Grade Explanations

If you want to request a regrade for an objective error on our part, you must submit a private request to the Instructors via Piazza; provide your UTLN and explain the error. The deadline for these requests is **one week** from the time the grades were posted for a given assignment; no exceptions to this policy will be made. A regrade request may or may not result in a new grade being assigned.

If a regrade request requires the course staff to make a manual modification to your submission, your grade for the component being regraded will be capped at a "Good" as a penalty. **IMPORTANT: If you misname a function, you could face a significant penalty from the autograder. Therefore it is crucial that you check that all function names match what the spec calls for exactly.**

You are always welcome to ask for an explanation of the grade you received. We want to help you understand how your work was evaluated and how you can continually improve in the course!

Collaboration and Academic Honesty

If you have **any** questions about the below policies or a specific situation dealing with academic integrity, do not hesitate to ask a course instructor. All students are expected to read and adhere to the Tufts Academic Integrity Policy.